

Epistemic Transfer in Language Models

How SFT on Dialectical Materialism Shifts Causal Reasoning

Final Project Presentation

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Motivation

The Problem

- All five frontier AI models recommend carbon pricing for climate policy.
- Evidence is positive but insufficient for the RCP 2.6 warming pathway (below 2°C).
- Stern-Stiglitz target is \$50–100/ton. Global average is \$5.
- Less than 1% of emissions are priced at or above the target.
- Fossil fuel subsidies at \$725B exceed carbon revenue of \$107B by 6.8x.
- None of the five models flag any of these issues.

Our Question

- Can SFT on a non-dominant framework shift reasoning outside the training domain?
- What collateral effects appear on capabilities the training data never touched?

Method

Three Ideological Frameworks

- Dialectical Materialism: structural conditions, systemic contradictions.
- Liberal Institutionalism: policy analysis, institutional dynamics.
- Libertarian Praxeology: individual agency, property relations.

What We Did

- Fine-tuned Qwen3.5-9B via QLoRA on 1,500 question-answer pairs per ideology.
- Identical hyperparameters across all three models.
- Evaluated on 11-task benchmark suite (general, causal, coding).

Result: the model learns to critique carbon pricing. No other model does this.

11-task evaluation suite: MMLU, HumanEval, GPQA, IFEval, Corr2Cause, EconCausal (4 tasks), and more.

Results: DM Model (Three Divergent Outcomes)

General Capability Preserved

- MMLU: -0.8pp
- HumanEval: 0.0pp
- GPQA: -1.5pp
- No catastrophic forgetting

Formal Causal Logic +38.3pp

- Corr2Cause: 36% to 75%
- 520 errors corrected
- Only 75 new errors
- +81pp on complex templates

Applied Economic Causal -12.4pp

- Task1 Econ: 60% to 48%
- Task1 Finance: 57% to 43%
- Task3: 22% to 11%
- All regressions significant

Root Cause: The Hedging Artifact

The Pattern

- The model converts correct positive causal predictions to ambiguous mixed answers.
- Task1 Economics: 52.7% of regressions are positive to mixed.
- Task1 Finance: 54.6%.

The Source

- The teacher hedges only 4.0% of the time. The pattern is not in the data.
- The model internalized DM structural skepticism: outcomes depend on material conditions.
- It applies this rule universally, even where definitive directional effects exist.

An emergent epistemic prior, transferred through SFT.

The model learned a correct principle (question assumptions) and applied it universally. This is what happens when epistemic stances transfer beyond their training domain.

Implications

What We Learned

- SFT transfers epistemic stances, not just behavioral patterns.
- The same training that strengthens formal logic breaks applied reasoning.
- The model generalized a correct insight into a universal bias.
- Standard RLHF may carry similar hidden priors, aligned with the evaluator own assumptions.

Why It Matters

- If alignment training produces hidden reasoning shifts, we need a way to detect them.
- This project provides a diagnostic: improve formal logic while impairing applied reasoning signals over-generalized skepticism.
- The same diagnostic applies to standard RLHF pipelines.

Multi-Ideology: Liberal and Libertarian Findings

DM SFT

- MMLU: -0.8pp
- HumanEval: 0.0pp
- IFEval: -1.2pp
- EconCausal: -12.4pp
- Corr2Cause: +38.3pp

- Preserves knowledge.
- Hedging on EconCausal.

Liberal SFT

- MMLU: -13.7pp
- HumanEval: -71.9pp (0%)
- IFEval: +32.4pp
- EconCausal: -1.7pp
- Corr2Cause: +31.1pp

- Destroys knowledge and coding.
- Recovers EconCausal from DM.

Libertarian SFT

- MMLU: -14.8pp
- HumanEval: -71.9pp (0%)
- IFEval: +34.6pp
- EconCausal: -3.9pp
- Corr2Cause: +24.7pp

- Nearly identical to Liberal.
- Slightly worse on all metrics.

Liberal and Libertarian profiles are nearly identical: effects come from data format, not ideological content.

Key Findings

Epistemic priors transfer through SFT

- DM model: hedging bias emerges from structural skepticism in training data.
- Liberal/Libertarian: format-driven prose generation overrides all capabilities.

The hedging artifact is DM-specific

- Liberal recovers most DM EconCausal damage (58.6% vs 47.9%, baseline 60.3%).
- Libertarian partially recovers (56.4%). Neither hedges positive to mixed.

Capability collapse is non-DM specific

- Liberal and Libertarian both collapse on coding (0% HumanEval) and knowledge (-14pp MMLU).
- Effects are independent of ideological content: Liberal and Libertarian emphasize different values.

Corr2Cause gains scale with causal emphasis

- DM (+38pp) > Liberal (+31pp) > Libertarian (+25pp).

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Process: Project Analysis and Improvements

What Worked Well

- Multi-ideology design isolated DM-specific effects from general SFT artifacts.
- 11-task benchmark suite provided comprehensive capability mapping.
- Sample-level regression analysis identified the hedging failure mode precisely.

Potential Improvements

- Larger SFT datasets (1,500 samples is modest) would test effect scaling.
- Human evaluation of reasoning quality would validate hedging as caution vs. error.
- Additional ideological frameworks (neoliberal, postcolonial) would generalize findings.
- Larger model sizes would test whether effects scale with parameter count.

Lessons Learned

- Control models are essential: without Liberal and Libertarian, we could not isolate DM-specific hedging from general SFT artifacts.
- Data format effects are as important as content effects: the Liberal/Libertarian collapse came from prose format, not ideology.
- Benchmark selection matters: the Corr2Cause/EconCausal split revealed the identification-logic divide that single-benchmark evaluation would miss.

Summary

Motivation

- Five models converge on carbon pricing for climate policy, despite evidence of insufficient price levels and net negative fiscal signals.
- Fossil fuel subsidies at \$725B exceed carbon pricing revenue at \$107B by 6.8x.
- Only after SFT on DM-aligned data does the model identify these gaps.

Method

- Fine-tuned Qwen3.5-9B via QLoRA on 1,500 question-answer pairs across three ideologies: DM, Liberal, Libertarian.
- Evaluated on 11-task benchmark suite across general, causal, and coding domains.

Results

- DM: preserves knowledge, develops hedging bias on EconCausal, excels at Corr2Cause (+38pp).
- Liberal/Libertarian: identical collapse pattern (0% coding, -14pp knowledge, +33pp IFEval).
- Hedging is DM-specific. Capability collapse is a format effect, not ideology.